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Section: C

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
image_path = "/content/Panda.png"

img = cv2.imread(image_path)

if img is None:
    print("Image not found. Please check the image path.")
else:
    print("Image loaded successfully!")
```

Image loaded successfully!

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(6, 5))
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

Original Grayscale Image



```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

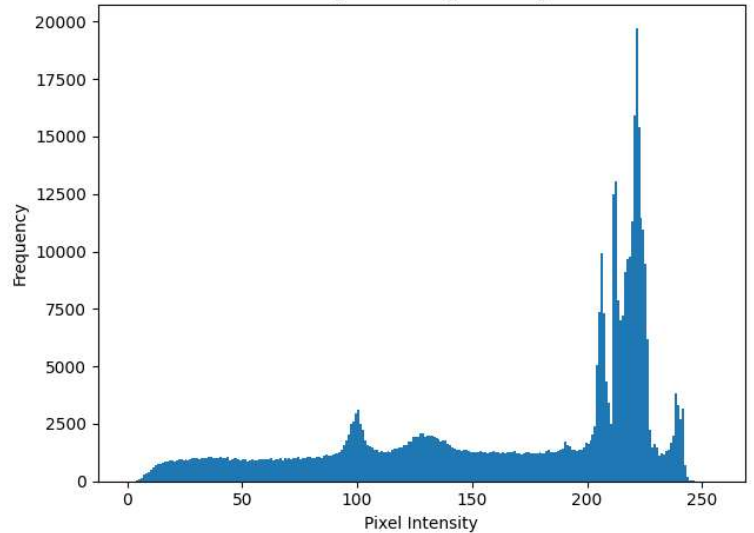
plt.subplot(1, 2, 2)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Histogram of Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

Original Grayscale Image



Histogram of Original Image



```
min_val = np.min(gray)
max_val = np.max(gray)

contrast_stretched = ((gray - min_val) / (max_val - min_val) * 255).astype(np.uint8)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched Image")
plt.axis("off")

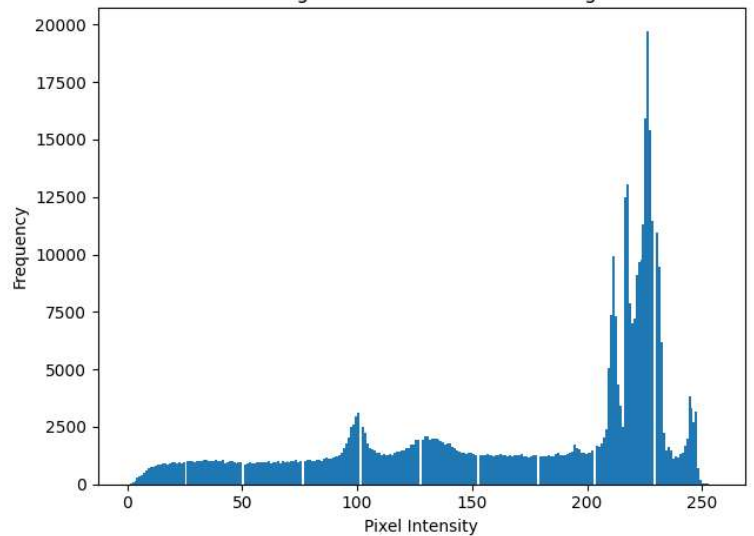
plt.subplot(1, 2, 2)
plt.hist(contrast_stretched.ravel(), bins=256, range=[0, 256])
plt.title("Histogram after Contrast Stretching")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

Contrast Stretched Image



Histogram after Contrast Stretching



```

hist_equalized = cv2.equalizeHist(gray)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalized Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(hist_equalized.ravel(), bins=256, range=[0, 256])
plt.title("Histogram after Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

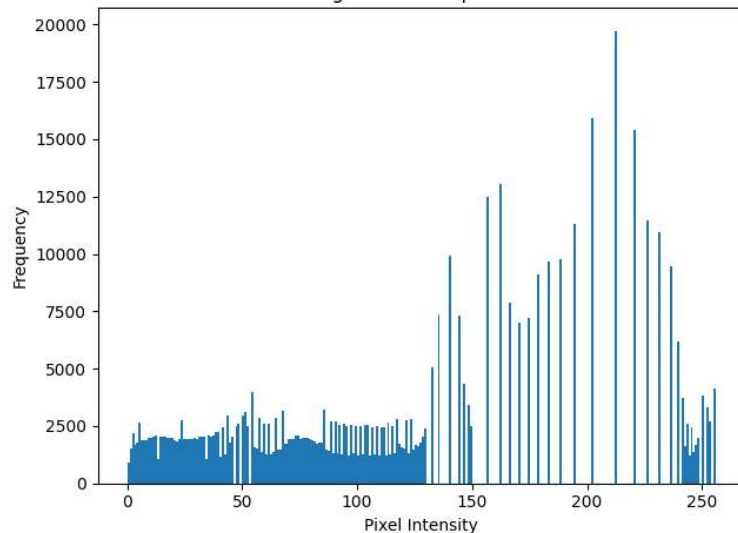
plt.tight_layout()
plt.show()

```

Histogram Equalized Image



Histogram after Equalization



```

clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))

clahe_image = clahe.apply(gray)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE Enhanced Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(clahe_image.ravel(), bins=256, range=[0, 256])
plt.title("Histogram after CLAHE")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

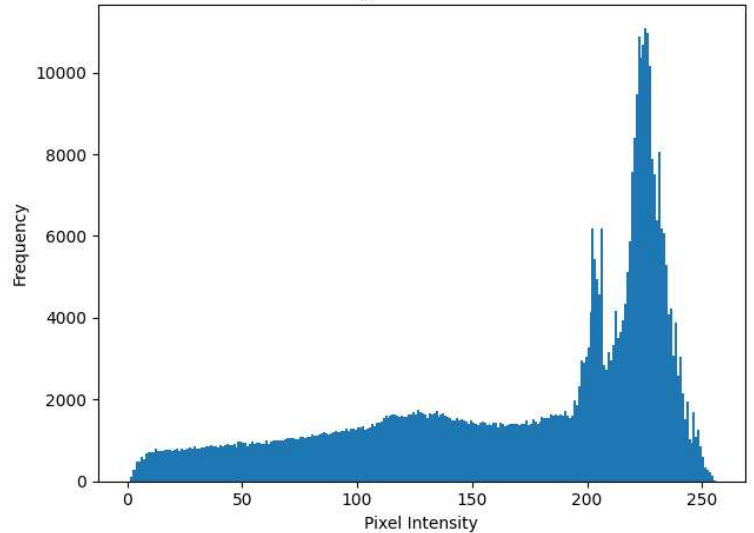
plt.tight_layout()
plt.show()

```

CLAHE Enhanced Image



Histogram after CLAHE



```
plt.figure(figsize=(16, 4))

plt.subplot(1, 4, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1, 4, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(1, 4, 3)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(1, 4, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original



Contrast Stretching



Histogram Equalization



CLAHE



```
plt.figure(figsize=(16, 4))

plt.subplot(1, 4, 1)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
```

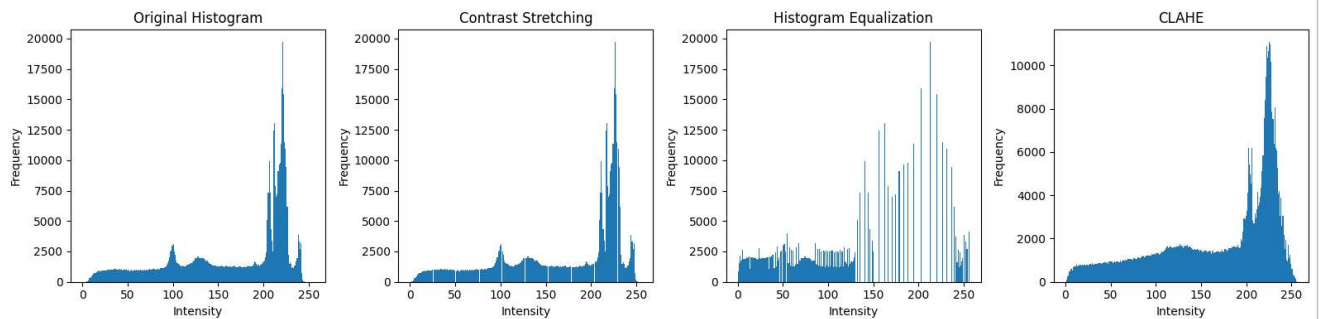
```
plt.title("Original Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 4, 2)
plt.hist(contrast_stretched.ravel(), bins=256, range=[0, 256])
plt.title("Contrast Stretching")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 4, 3)
plt.hist(hist_equalized.ravel(), bins=256, range=[0, 256])
plt.title("Histogram Equalization")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 4, 4)
plt.hist(clahe_image.ravel(), bins=256, range=[0, 256])
plt.title("CLAHE")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(14, 8))

plt.subplot(2, 4, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(2, 4, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 4, 3)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(2, 4, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

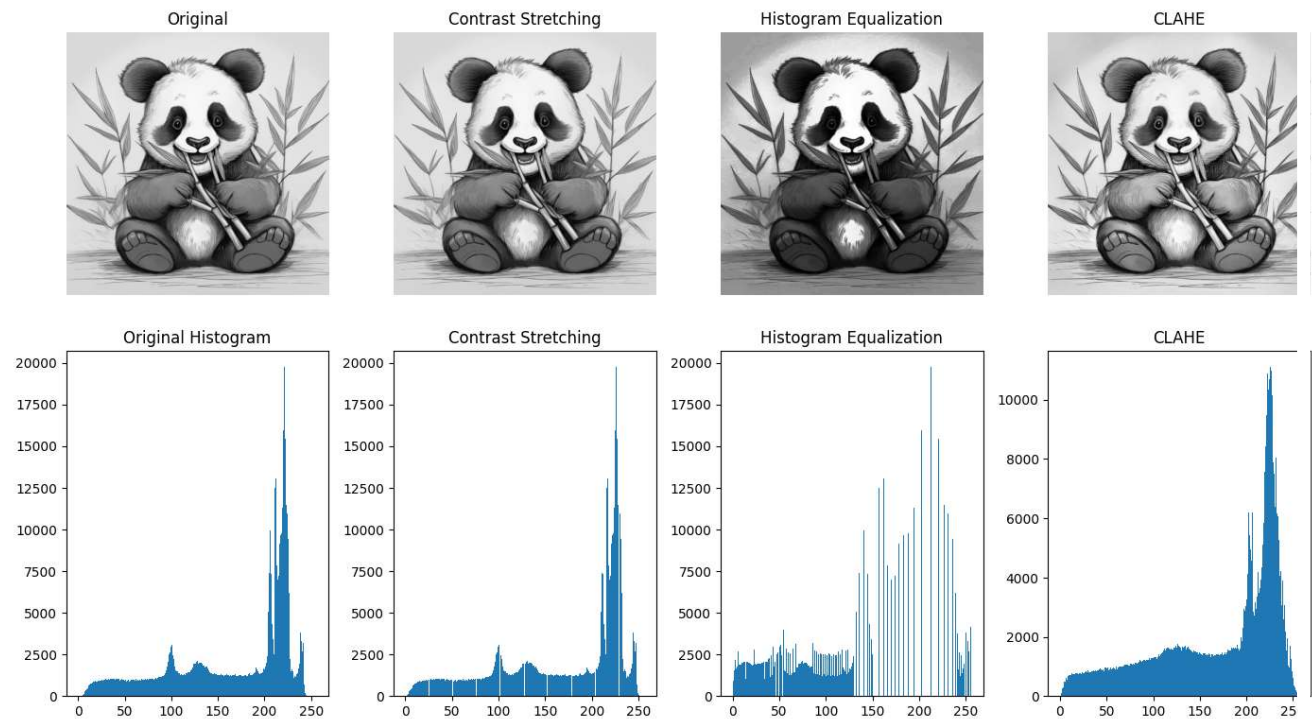
plt.subplot(2, 4, 5)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")

plt.subplot(2, 4, 6)
plt.hist(contrast_stretched.ravel(), bins=256, range=[0, 256])
plt.title("Contrast Stretching")

plt.subplot(2, 4, 7)
plt.hist(hist_equalized.ravel(), bins=256, range=[0, 256])
plt.title("Histogram Equalization")
```

```
plt.subplot(2, 4, 8)
plt.hist(cclahe_image.ravel(), bins=256, range=[0, 256])
plt.title("CLAHE")

plt.tight_layout()
plt.show()
```



```
cv2.imwrite("/content/contrast_stretched.png", contrast_stretched)
cv2.imwrite("/content/histogram_equalized.png", hist_equalized)
cv2.imwrite("/content/cclahe_image.png", cclahe_image)

print("All enhanced images saved successfully!")
```

All enhanced images saved successfully!

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